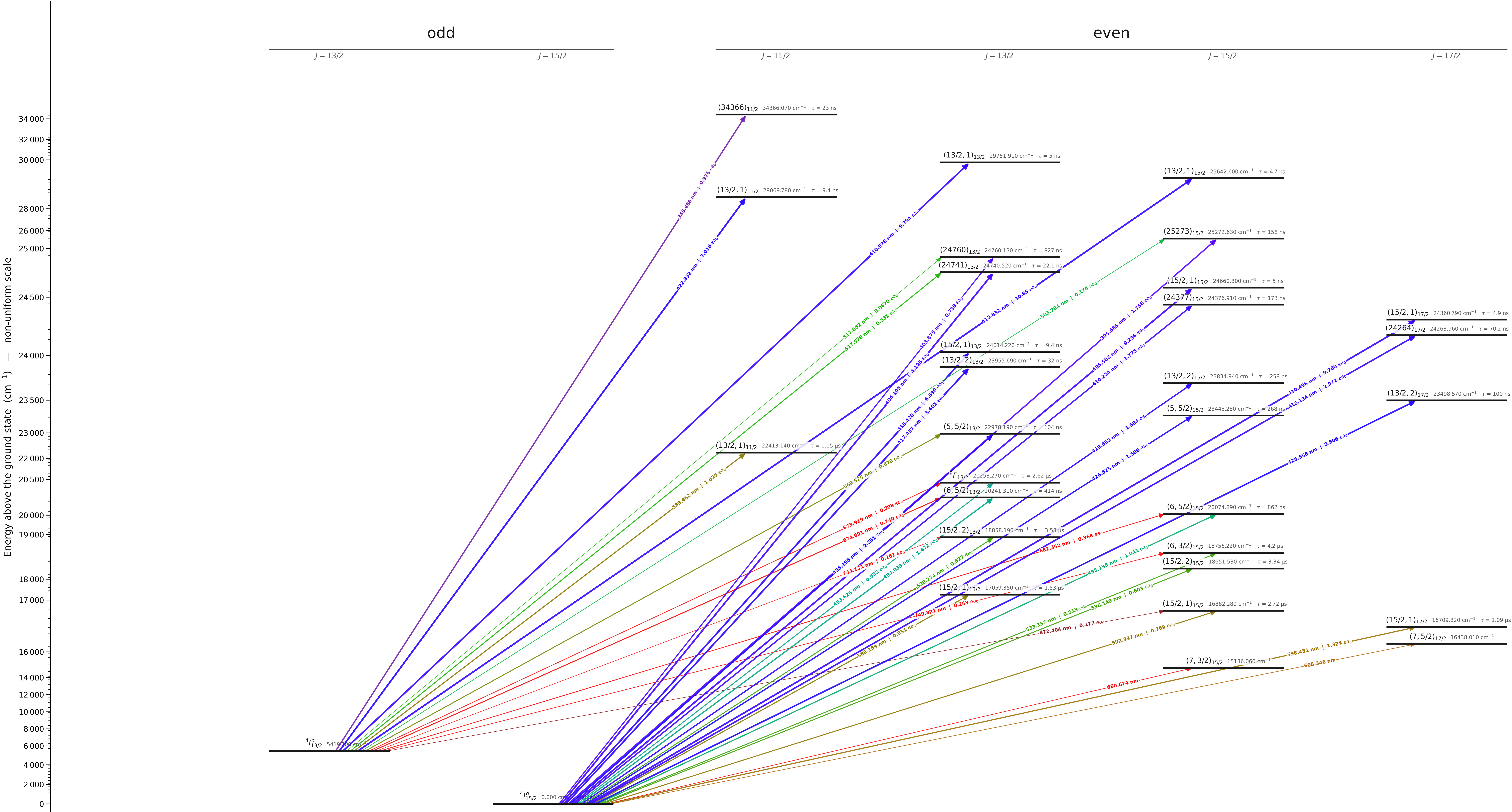
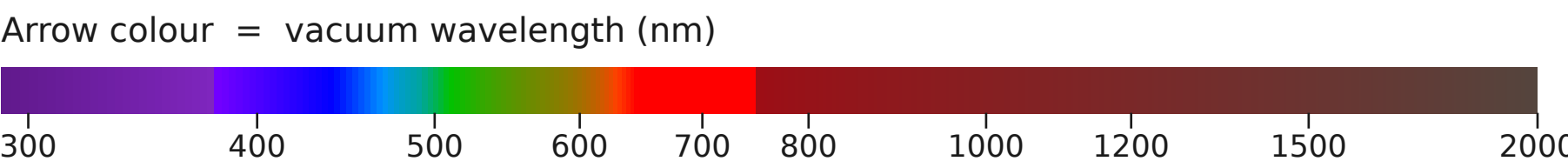


# Holmium

Ho (Ho I). Strongest classified transitions below 2 μm (NIST): 31 levels, 39 lines.

Ho<sup>+</sup> ionisation limit 48567.00 cm<sup>-1</sup> = 6.02154 eV (205.901 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|U||er||U'|$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                     | λ vac (nm) | λ air (nm) | ν (THz)     | ⟨U er U'⟩  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                  |
|----------------------------------------------------------------|------------|------------|-------------|-------------------------------|----------------------|----------------|------------|------------------------------------------------------|
| <sup>4</sup> I <sub>3/2</sub> - (34366) <sub>11/2</sub>        | 345.4665   | 345.3675   | 867.7903    | 0.976                         | 3.900e+06            | 6.92           | 8.97       | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>5/2</sub> - (25273) <sub>15/2</sub>        | 395.6850   | 395.5730   | 757.6544    | 1.756                         | 6.300e+06            | 1.007          | 99.5       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (24760) <sub>13/2</sub>        | 403.8751   | 403.7610   | 742.2900    | 0.739                         | 1.200e+06            | 0.1924         | 99.2       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (24741) <sub>13/2</sub>        | 404.1952   | 404.0811   | 741.7021    | 4.125                         | 3.730e+07            | 7.202          | 82.4       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (15/2, 1) <sub>15/2</sub>      | 405.5018   | 405.3873   | 739.3122    | 9.236                         | 1.620e+08            | 31.83          | 81         | strong blue line                                     |
| <sup>4</sup> I <sub>5/2</sub> - (24377) <sub>15/2</sub>        | 410.2243   | 410.1085   | 730.8014    | 1.775                         | 5.780e+06            | 0.92           | 100        | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (15/2, 1) <sub>17/2</sub>      | 410.4957   | 410.3799   | 730.3181(3) | 9.760                         | 1.550e+08            | 32.48          | 75.9       | main broad cooling line: MOT, Zeeman slower, imaging |
| <sup>4</sup> I <sub>3/2</sub> - (13/2, 1) <sub>13/2</sub>      | 410.9779   | 410.8619   | 729.4613    | 9.794                         | 2.000e+08            | 31.83          | 100        | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>5/2</sub> - (24264) <sub>17/2</sub>        | 412.1339   | 412.0176   | 727.4152    | 2.972                         | 1.420e+07            | 2.267          | 99.7       | cycling line 412.1 nm, two-electron jump             |
| <sup>4</sup> I <sub>5/2</sub> - (13/2, 1) <sub>15/2</sub>      | 412.8325   | 412.7161   | 726.1843    | 10.85                         | 2.120e+08            | 33.86          | 99.6       | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>5/2</sub> - (15/2, 1) <sub>13/2</sub>      | 416.4199   | 416.3026   | 719.9282    | 6.690                         | 8.970e+07            | 16.93          | 84.3       | strong blue line                                     |
| <sup>4</sup> I <sub>5/2</sub> - (13/2, 2) <sub>13/2</sub>      | 417.4374   | 417.3197   | 718.1735    | 3.601                         | 2.580e+07            | 4.974          | 82.6       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (13/2, 2) <sub>15/2</sub>      | 419.5521   | 419.4339   | 714.5535    | 1.504                         | 3.880e+06            | 0.6169         | 100        | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (13/2, 1) <sub>11/2</sub>      | 422.8316   | 422.7125   | 709.0116    | 7.018                         | 1.100e+08            | 16.93          | 100        | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>5/2</sub> - (13/2, 2) <sub>17/2</sub>      | 425.5578   | 425.4380   | 704.4694    | 2.806                         | 1.150e+07            | 1.592          | 100        | cycling line 425.6 nm                                |
| <sup>4</sup> I <sub>5/2</sub> - (5, 5/2) <sub>15/2</sub>       | 426.5251   | 426.4051   | 702.8718    | 1.506                         | 3.700e+06            | 0.5939         | 99.2       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (5, 5/2) <sub>13/2</sub>       | 435.1953   | 435.0730   | 688.8688    | 2.251                         | 8.900e+06            | 1.53           | 92.6       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - <sup>8</sup> F <sub>13/2</sub> | 493.6256   | 493.4878   | 607.3277    | 0.532                         | 3.400e+05            | 0.06075        | 89.1       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (6, 5/2) <sub>13/2</sub>       | 494.0392   | 493.9013   | 606.8192    | 1.472                         | 2.600e+06            | 0.3844         | 100        | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (6, 5/2) <sub>15/2</sub>       | 498.1347   | 497.9958   | 601.8301    | 1.041                         | 1.110e+06            | 0.1846         | 95.7       | ground-state line                                    |
| <sup>4</sup> I <sub>5/2</sub> - (25273) <sub>15/2</sub>        | 503.7040   | 503.5636   | 595.1759    | 0.174                         | 2.990e+04            | 1.007          | 0.47       | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>3/2</sub> - (24760) <sub>13/2</sub>        | 517.0516   | 516.9076   | 579.8115    | 0.067                         | 4.700e+03            | 0.1924         | 0.39       | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>3/2</sub> - (24741) <sub>13/2</sub>        | 517.5764   | 517.4323   | 579.2236    | 0.581                         | 3.520e+05            | 7.202          | 0.78       | line from the metastable 4f11 6s2 4lo13/2 level      |
| <sup>4</sup> I <sub>5/2</sub> - (15/2, 2) <sub>13/2</sub>      | 530.2736   | 530.1261   | 565.3543    | 0.527                         | 2.700e+05            | 0.04446        | 96.7       | ground-state line                                    |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2τ).

## <sup>165</sup>Ho hyperfine structure (I = 7/2)

| Level                         | A (MHz)       | B (MHz)       | Shift of each F level from the centre of gravity (MHz)                                                                  |
|-------------------------------|---------------|---------------|-------------------------------------------------------------------------------------------------------------------------|
| <sup>4</sup> I <sub>5/2</sub> | 800.583645(6) | -1668.0053(3) | F=4: -24424.99 F=5: -20115.71 F=6: -15019.45 F=7: -9177.08 F=8: -2636.25 F=9: +4548.58 F=10: +12316.13 F=11: +20598.32  |
| (15/2, 1) <sub>17/2</sub>     | 654.9(3)      | -620(20)      | F=5: -21991.97 F=6: -17945.34 F=7: -13253.91 F=8: -7931.35 F=9: -1993.29 F=10: +4542.69 F=11: +11657.04 F=12: +19328.28 |

Measured A, B constants; sources in the data file.